

TCC805x MCU BSP

API Specification for Windowed Watchdog Timer

Rev. 1.01 [A]

2021-01-22

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1 INTRODUCTION

This document describes the windowed watchdog timer device driver (hereinafter referred to as WDT driver) of TCC805x MCU BSP. For more detailed information of the windowed watchdog timer in the TCC805x MCU Sub-system, refer to Chapter 20 Windowed Watchdog Timer (WWDT) of the ***"PART 11-MICOM SUB-SYSTEM"*** in the ***"TCC805x Full Specification"***. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

WWDT	Windowed Watchdog Timer
-------------	-------------------------

3 SOURCE FILES

3.1 Location of Source Files

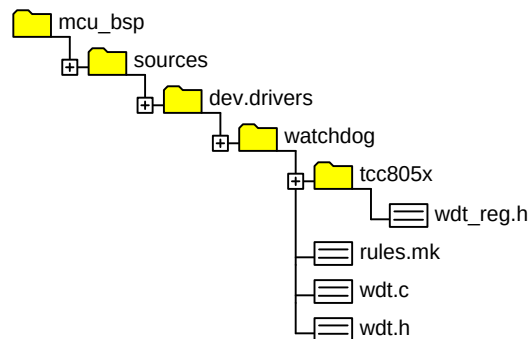


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- wdt_reg.h
 - Device register address and writing password that are dependent on chipset
- rules.mk
 - Rules used for build
- wdt.c
 - WDT driver source
- wdt.h
 - WDT driver API header

4 ENUMERATIONS

4.1 WDTResetType_t

Enumeration variables to select whether handshake mode with FMU is enabled or not

Declaration

```
typedef enum WDTResetType
{
    WDT_RST_DIRECT      = 0x0U,
    WDT_RST_FMU_HANDLER = 0x4U
} WDTResetType_t;
```

Description

Value	Description
WDT_RST_DIRECT	<i>The system is reset without any interrupting.</i>
WDT_RST_FMU_HANDLER	<i>Activated RST_REQ signal is treated by Fault Management Unit (FMU). You can handle the process after Reset signal.</i>

Remark

None

4.2 WDT2oo3Type_t

Enumeration variables to select one of RST counter timeout signals or filtered signal which is 2 out of 3 voted.

Declaration

```
typedef enum WDT2003Type
{
    WDT_2003_VOTE = 0x0U,
    WDT_ZERO_0    = 0x1U,
    WDT_ZERO_1    = 0x2U,
    WDT_ZERO_2    = 0x3U
} WDT2oo3Type_t;
```

Description

Value	Description
WDT_2003_VOTE	Watchdog interrupt and watchdog reset request signal select 2 out of 3 vote applied signal.
WDT_ZERO_0	IRQ_CNT_ZERO_0 and RST_CNT_ZERO_0.
WDT_ZERO_1	IRQ_CNT_ZERO_1 and RST_CNT_ZERO_1.
WDT_ZERO_2	IRQ_CNT_ZERO_2 and RST_CNT_ZERO_2.

Remark

None

4.3 WDTErrorCode_t

Enumeration variables to select one of RST counter timeout signals or filtered signal which is 2 out of 3 voted.

Declaration

```
enum
{
    WDT_ERR_ALREADY_DONE      = (uint32)SAL_DRVID_WDT,
    WDT_ERR_NOT_INITIALIZED   = (uint32)SAL_DRVID_WDT + 1UL,
    WDT_ERR_INVALID_PARAM     = (uint32)SAL_DRVID_WDT + 2UL,
    WDT_ERR_NOT_SUPPORT_MODE  = (uint32)SAL_DRVID_WDT + 3UL
}
```

Description

Value	Description
WDT_ERR_ALREADY_DONE	A job is re-executed.
WDT_ERR_NOT_INITIALIZED	WDT module is <u>not</u> initialized.
WDT_ERR_INVALID_PARAM	The invalid parameters are passed.
WDT_ERR_NOT_SUPPORT_MODE	The mode is <u>not</u> supported.

Remark

None

5 CONSTANTS

5.1 Function Indexes

The indexing constants correspond with each API function. These indexes are used for debugging or for informing the information of WDT API function.

Declaration

```
#define WDT_API_COMMON_INDEX (0x0)
#define WDT_API_STOP         (WDT_API_COMMON_INDEX + 0)
#define WDT_API_SET_SM_MODE  (WDT_API_COMMON_INDEX + 1)
#define WDT_API_KICK_ACT     (WDT_API_COMMON_INDEX + 2)
#define WDT_API_RESET_SYS    (WDT_API_COMMON_INDEX + 3)
#define WDT_API_INIT         (WDT_API_COMMON_INDEX + 4)
```

Description

<i>Value</i>	<i>Description</i>
WDT_API_COMMON_INDEX	<i>The starting index of timer module</i>
WDT_API_STOP	<i>The index of WDT_Stop() function</i>
WDT_API_SET_SM_MODE	<i>The index of WDT_SmMode() function</i>
WDT_API_KICK_ACT	<i>The index of WDT_KickPing() function</i>
WDT_API_RESET_SYS	<i>The index of WDT_ResetSystem() function</i>
WDT_API_INIT	<i>The index of WDT_Init() function</i>

Remark

None

6 APIs

6.1 WDT_KickPing

Processes kick action to counter of WDT as well as clears and resets IRQ counter.

Declaration

```
void WDT_KickPing  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.2 WDT_ResetSystem

Resets or reboots the system right away.

Declaration

```
void WDT_ResetSystem  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.3 WDT_Stop

Stops WDT counter and unregister interrupt handler.

Declaration

```
void WDT_Stop  
(  
    void  
);
```

Parameters

None

Return

None

Remark

None

6.4 WDT_SmMode

Sets configuration of safety mechanism on WDT device.

Declaration

```
SALRetCode_t WDT_SmMode
(
    boolean          ucUseFmu,
    WDTResetType_t   uiUseHandler,
    WDT2oo3Type_t     uiVote
);
```

Parameters

Value	Description
<i>ucUseFmu</i>	<i>[in] The flag to use handshake with FMU. If true, handshake mode will be enabled.</i>
<i>uiUseHandler</i>	<i>[in] When handshake with FMU, external FMU handler is available. Refer to the WDTResetType t.</i>
<i>uiVote</i>	<i>[in] WDT reset request selection mode, selecting one of 3 reset request line or 2 out of 3 voted reset requests. Refer to the WDT2oo3Type t.</i>

Return

Value	Description
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

6.5 WDT_Init

Initializes configuration registers, registers interrupt handler for kick action, and starts WDT counter.

Declaration

```
SALRetCode_t WDT_Init  
(  
    void  
);
```

Parameters

None

Return

<i>Value</i>	<i>Description</i>
<i>SAL_RET_SUCCESS</i>	<i>Success</i>
<i>SAL_RET_FAILED</i>	<i>Failure</i>

Remark

None

7 HOW TO USE WDT DRIVER

7.1 Default Operation

The normal operation of Watchdog is turned on by MCU system via ***WDT_Init()***. If the operation of Watchdog is off via ***WDT_Stop()***, MCU system reboots after configuring reset counter value.

```
{
    SALRetCode_t ret;

    ret = WDT_Init();

    if (ret == SAL_RET_SUCCESS)
    {
        ...
    }
}
```

```
{
    WDT_Stop();
}
```

7.2 Customized Operation

You can customize the operation by using Watchdog driver and Timer driver. You should call the ***WDT_KickPing()*** for MCU system to be alive.

```
#define WDT_IRQ_TIME          (10UL) // 10 seconds
#define WDT_RESET_TIME       (24UL) // 24 seconds
#define TIMER_TESTING_PRERIGHT_TIME (10 * 1000 * 1000) // 10 sec

{
    SALRetCode_t ret;

    (void)WDT_SmMode(TRUE, WDT_RST_DIRECT, WDT_2003_VOTE);

    ret = WDT_SetTimeout(WDT_RESET_TIME, WDT_IRQ_TIME);

    if (ret == SAL_RET_SUCCESS)
    {
        (void)TIMER_EnableWithMode(TIMER_CH_1,
                                    TIMER_TESTING_PRERIGHT_TIME,
                                    TIMER_OP_FREERUN,
                                    TIMER_Handler,
                                    NULL_PTR);

        GIC_IntSrcEn(GIC_TIMER_0 + TIMER_CH_1);

        WDT_Start();
    }
}
```

```
sint32 TIMER_Handler(TIMERChannel_t iChannel, void * pArgs)
{
    WDT_KickPing();

    return SAL_RET_SUCCESS;
}
```

8 REFERENCES

- [1] TCC805x Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 1.01: 2021-01-22

- Updated
 - Chapter 4.3: Declaration

Rev. 1.00: 2020-11-12

- First version release

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